EPSR-test-ex106 Pressure control through the repulsive cores (control-p p)

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the `p` variant of `control-p` in the `epsr` engine [1,2,3]: instead of changing the box, the minimum distance `rminex` of every pair with an exponential repulsive core is scaled after each iteration, which changes the effective excluded volume and so the pressure at fixed density.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (pressure block, `pfac`), `multiply_pcoefex` (scales `rminex`, rewrites the pair symbol)
- `epsr_standalone_rev3.f`: `generate_rp` (exponential repulsive core), `repulsivept` (`rminex` > `0` branches)
- `ato_manipulation_rev4.f`: `press`; `read_and_save_files.f`: `read_pcof`, `save_files_others`

Rust (port, rust/epsr-core/src/):

- `pressure.rs`: `control_pressure`, `multiply_pcoefex`
- `potential.rs`: `generate_rp`, `repulsivept`; `energy.rs`: `press`
- `pcof.rs`: `read_pcof`, `write_pcof_pairs`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. For this case two pairs carry a repulsive core (`rminex pcoefex coordex`): DMSO O - water O1 `2.6 0.5 0` and O1-O1 `2.7 0 0`. After each iteration `pfac = 1/(1+0.01*clip(pressd))` multiplies `rminex` of every eligible pair and the reference potential is regenerated.

Inputs

`scripts/make_inputs.sh`: `.EPSR.inp` with `control-p p 0.0 0.01`, `ntimes 2`, `niter 2`; `.pcof` with the two core lines above (all other pairs 0 as in EPSR-test). `commands.txt`: `epsr 200DMSO400Water.EPSR.inp`, `repeat 2`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
The feature is active: in the stored Fortran reference `.pcof` after four iterations O-O1 has `rminex` 2.4985 and O1-O1 2.5946, and the O1-O1 amplitude has been grown from 0 to 0.038 by `repulsivept`. The README reports gcov evidence: `multiply_pcoefex` is called in every iteration and the `rminex` multiplication executes 8 times (the one-word branch of the symbol rewrite is not reached).

Notes

The pair symbol is rewritten with a list-directed write of `rminex` followed by the remaining words of the old value, as in the Fortran.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.